WINDOWS ENDPOINT THREAT DETECTION AND INVESTIGATION

Using Wazuh SIEM and Sysmon

Hands-on SOC Analyst L1 Lab Project

VMware Workstation Pro • Windows 11 endpoint • Wazuh SIEM
Evidence collected: 16 October 2026

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1. Executive summary

This project builds a small, isolated security monitoring lab. A Windows 11 virtual machine named SOCWIN11 runs Sysmon and the Wazuh agent. Sysmon records detailed endpoint activity, including process creation and termination. The Wazuh agent forwards Windows event-channel records to a separate Wazuh server, where events can be searched and reviewed in Threat Hunting.

The screenshots show local Sysmon Event ID 1 and Event ID 5 records, events from SOCWIN11 in Wazuh, and Sysmon-specific fields in a Wazuh event document. The evidence also includes Wazuh detections mapped to MITRE ATT&CK.; Alerts are leads for investigation; they do not by themselves prove compromise.

Status summary

• Windows endpoint: Windows 11 VM, hostname SOCWIN11; 8 GB RAM, 2 vCPU, 64 GB disk, NAT.

• Wazuh server: Amazon Linux 2023, Wazuh 4.14.8, hostname wazuh-server; IP shown as 192.168.219.130.

• Endpoint IP shown: 192.168.219.131; Wazuh agent 4.14.8, agent ID 003 in event details.

• Sysmon installed and running; local Event IDs 1 and 5 observed.

• Sysmon forwarding confirmed by Wazuh fields for Microsoft-Windows-Sysmon/Operational and Event ID 1.

2. Objectives and scope

• Monitor a dedicated Windows endpoint VM, not the host computer.

• Install Sysmon and collect process telemetry.

• Connect the Wazuh agent to the existing Wazuh server.

• Forward the Sysmon Operational channel to Wazuh.

• Search events, inspect rule metadata, and practice SOC L1 triage.

Scope note: This is a controlled learning lab. No claim is made that malware was executed or that the endpoint was compromised.

3. Lab architecture and environment

Event flow: Windows activity → Sysmon Operational log → Wazuh agent on SOCWIN11 → Wazuh manager/indexer → Threat Hunting search and alert review.

• Host: Windows 11 Pro, 16 GB RAM, runs VMware.

• Endpoint: Windows 11 VM SOCWIN11, 8 GB RAM, 2 vCPU, 64 GB disk, NAT, IP 192.168.219.131 (as captured).

• Server: Amazon Linux 2023, Wazuh 4.14.8, wazuh-server, IP 192.168.219.130 (as captured).

IP addresses can change after a VM restart if assigned dynamically.

4. Implementation completed

4.1 Windows endpoint

The Windows 11 endpoint VM was used for the lab. The guest hostname is SOCWIN11; endpoint collection is scoped to the VM, not the host.

4.2 Sysmon

Sysmon is installed and running. Event Viewer → Applications and Services Logs → Microsoft → Windows → Sysmon → Operational shows Event ID 1 (Process Create) and Event ID 5 (Process Terminated).

4.3 Wazuh agent

The agent is installed on SOCWIN11. Wazuh event documents show agent.name SOCWIN11 and agent.id 003.

4.4 Sysmon forwarding

The agent configuration includes a localfile entry for Microsoft-Windows-Sysmon/Operational. A Wazuh document contains data.win.system.channel = Microsoft-Windows-Sysmon/Operational and eventID = 1, confirming that Sysmon telemetry reached the SIEM.

4.5 Event search

Threat Hunting Events displays endpoint records, rule IDs, descriptions, event fields, and MITRE ATT&CK; metadata.

5. Evidence and screenshot walkthrough

The following figures are screenshots supplied during the lab. Adjacent screenshots may show different scroll positions in the same event details.

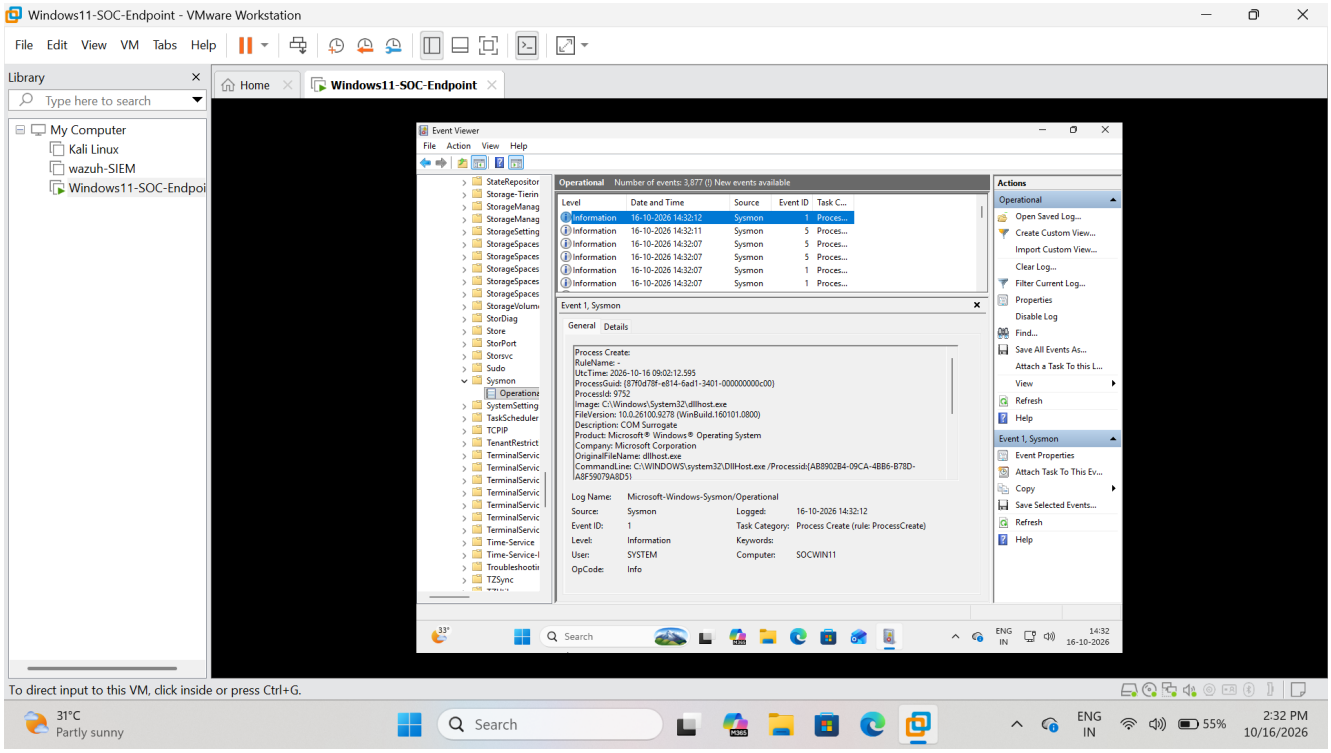


Figure 1. Sysmon Event ID 1 — Process Create in Windows Event Viewer

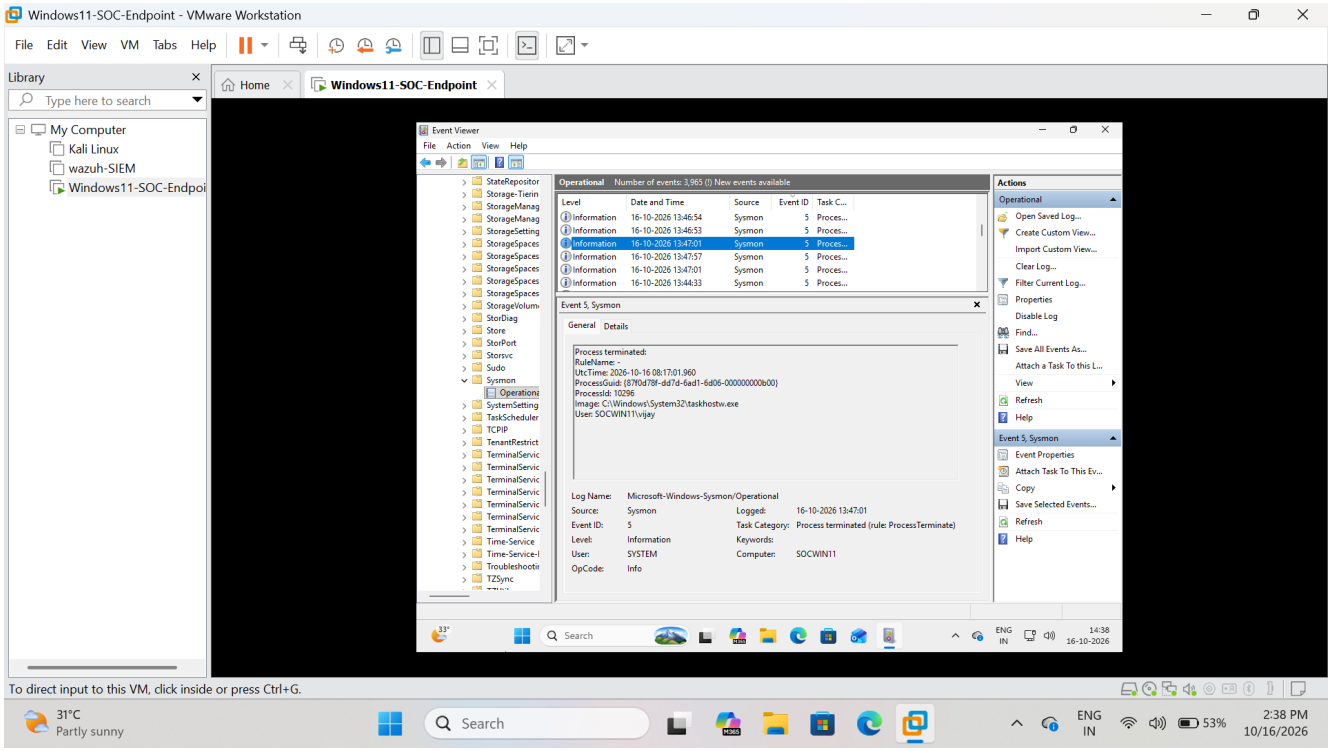


Figure 2. Sysmon Event ID 5 — Process Terminate in Windows Event Viewer

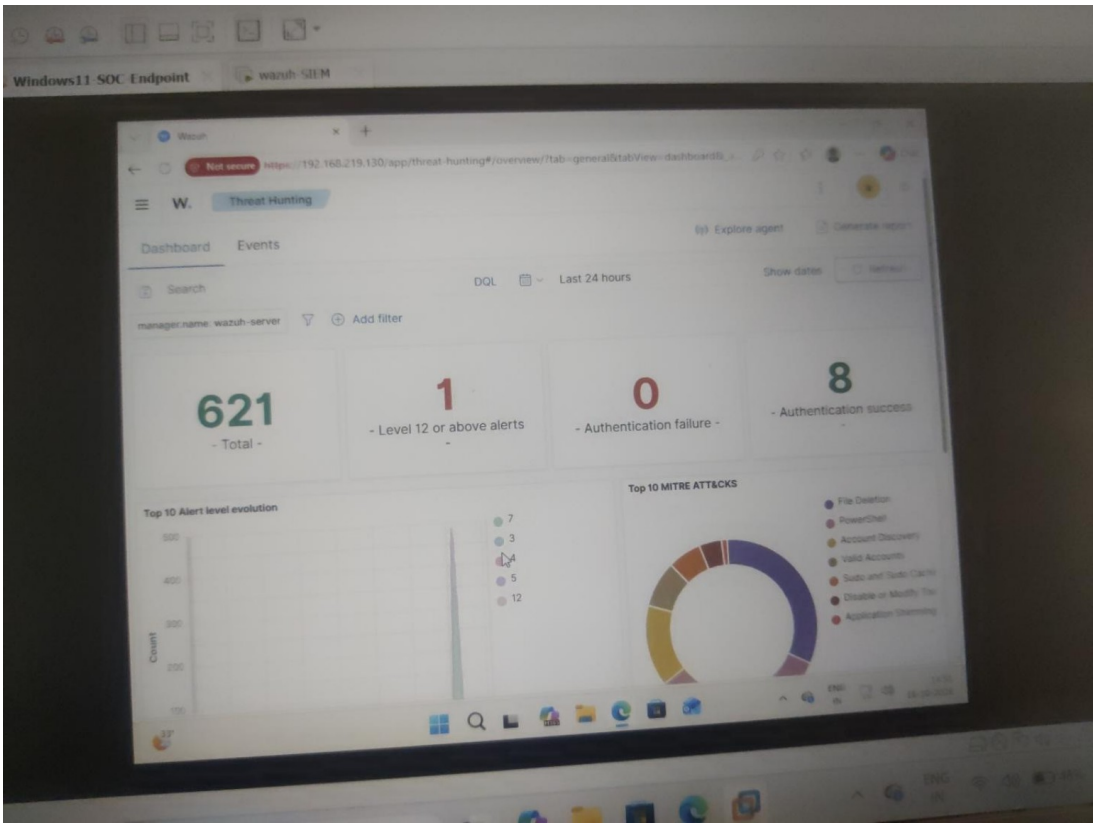


Figure 3. Wazuh Threat Hunting overview dashboard (captured earlier)

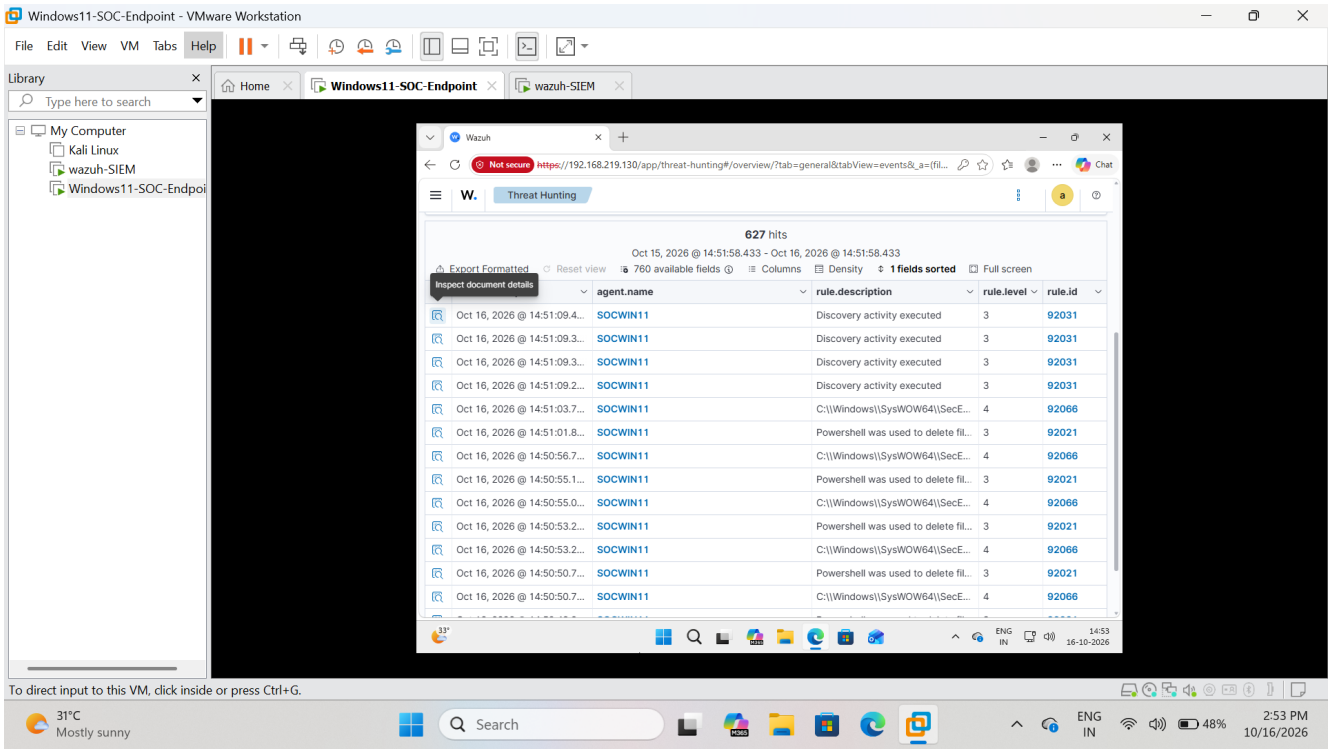


Figure 4. Wazuh Threat Hunting event table showing events from SOCWIN11

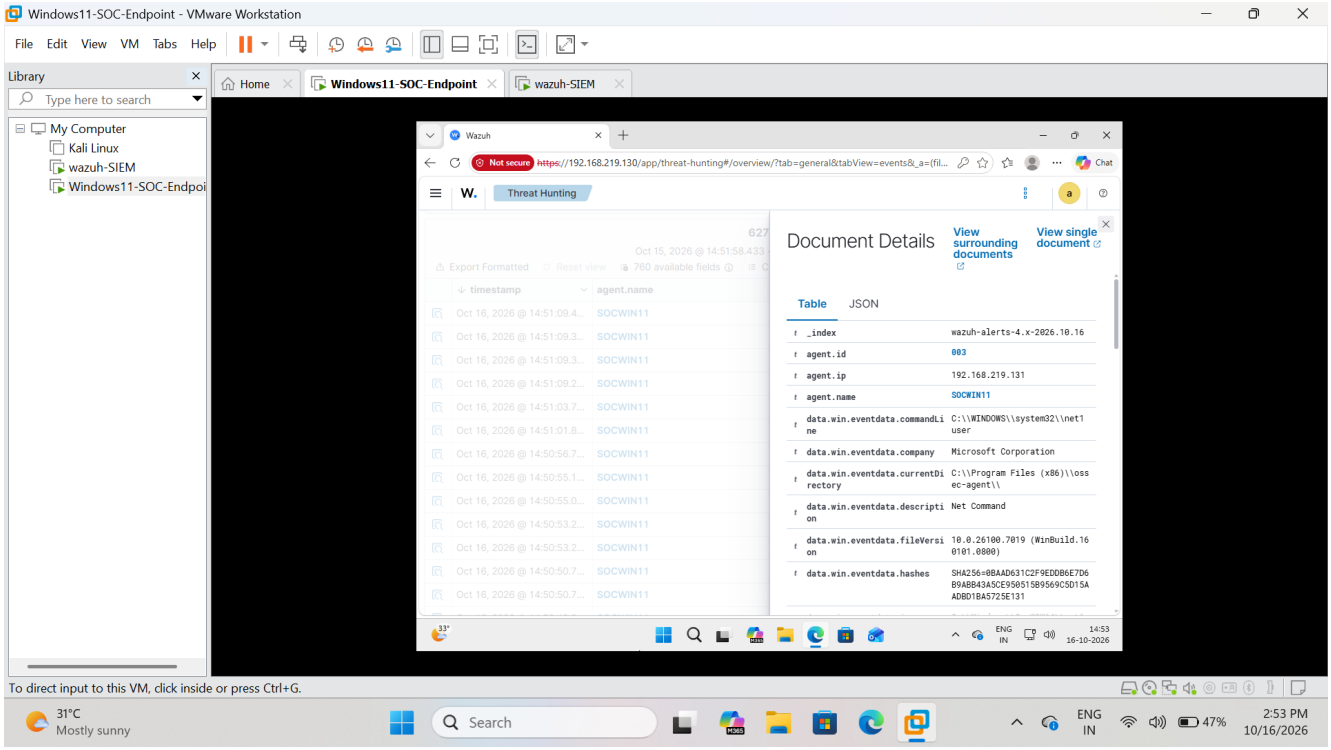


Figure 5. Wazuh document details — endpoint command-line telemetry

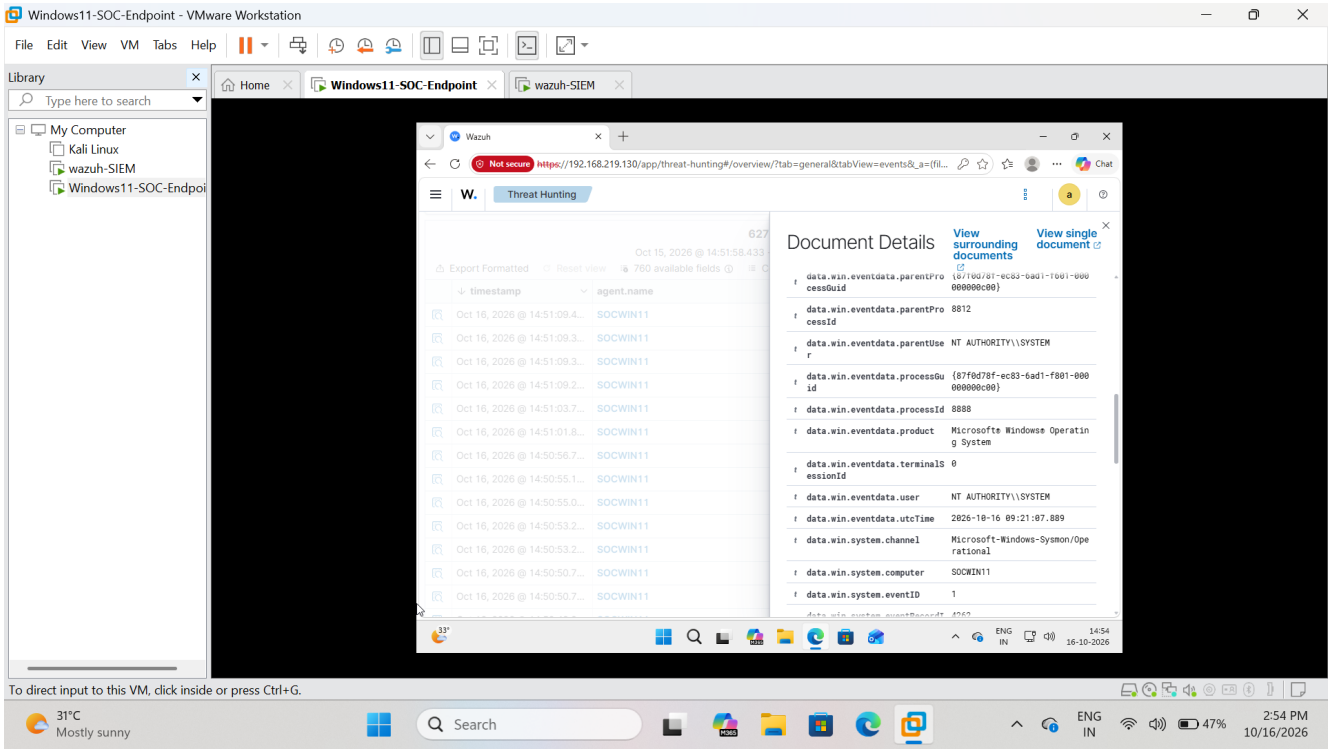


Figure 6. Wazuh document details — Sysmon channel and Event ID 1

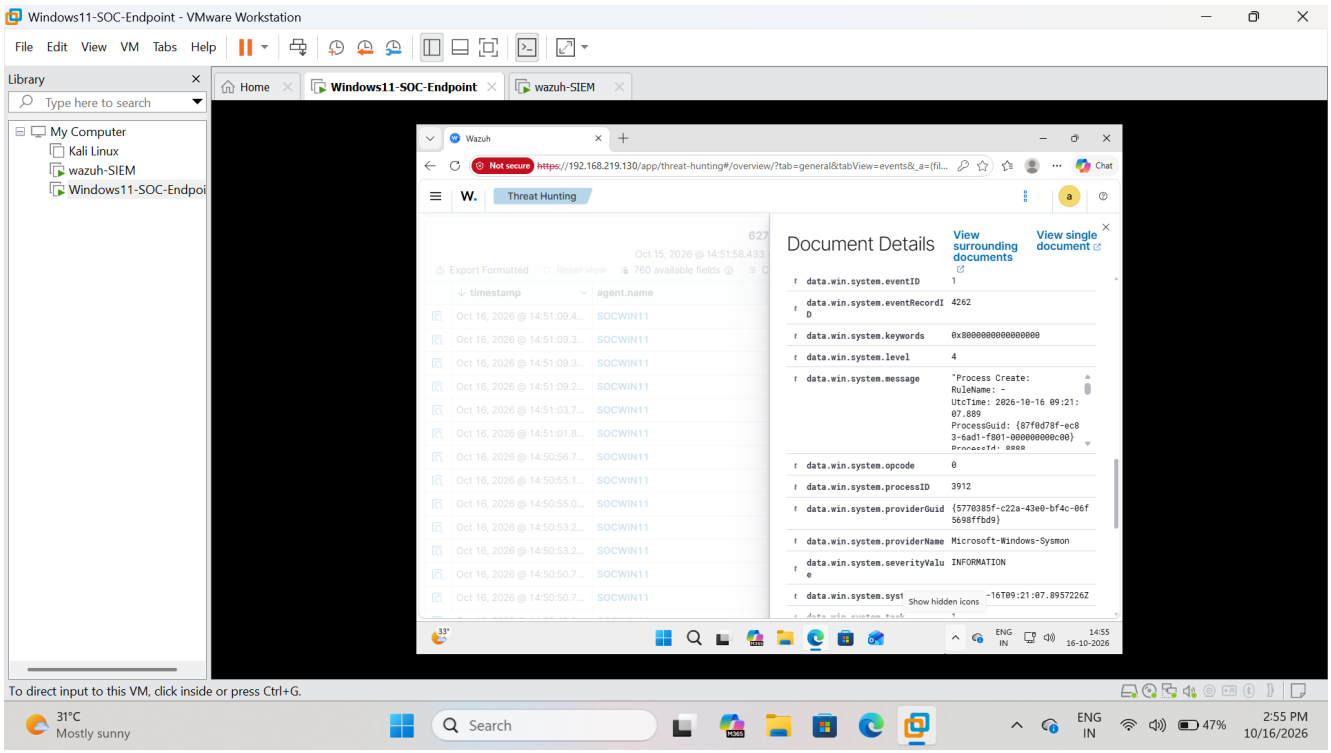


Figure 7. Wazuh detection details — Rule 92066 and MITRE PowerShell mapping

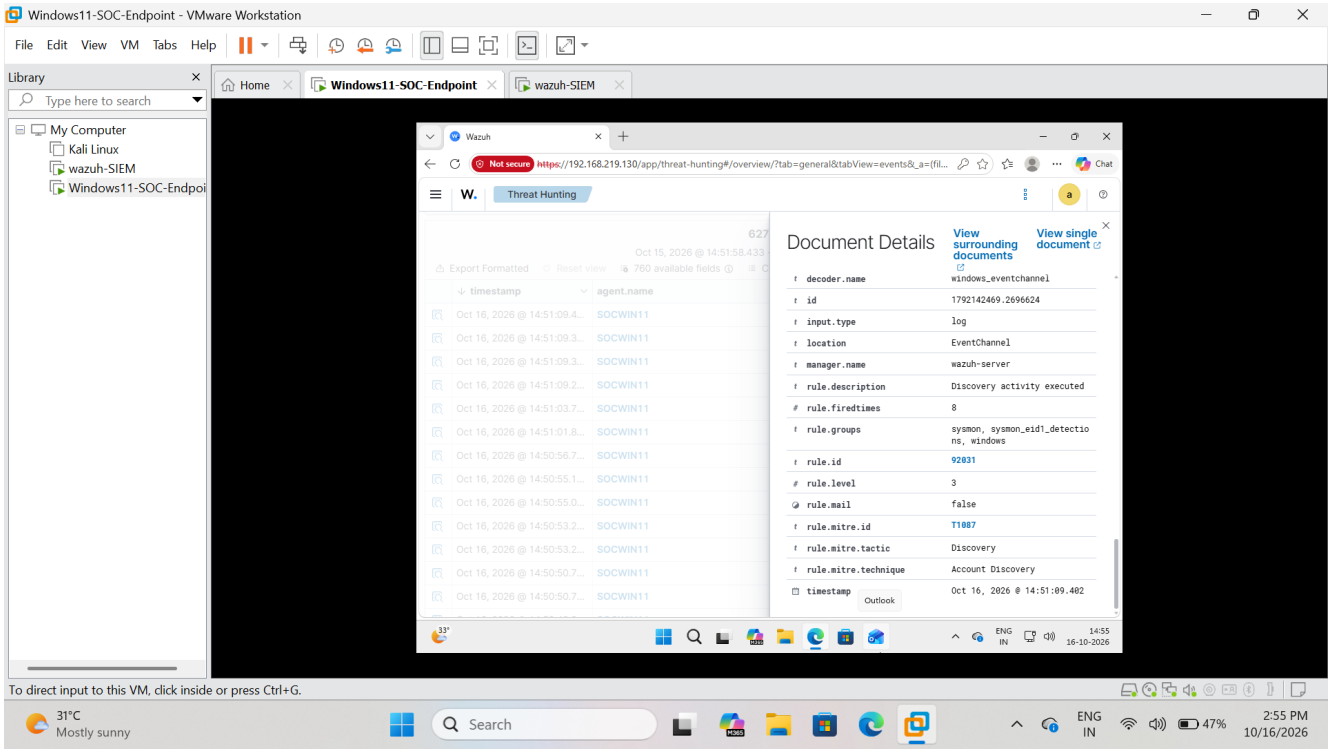


Figure 8. Wazuh event details — SOCWIN11 agent identity and Windows fields

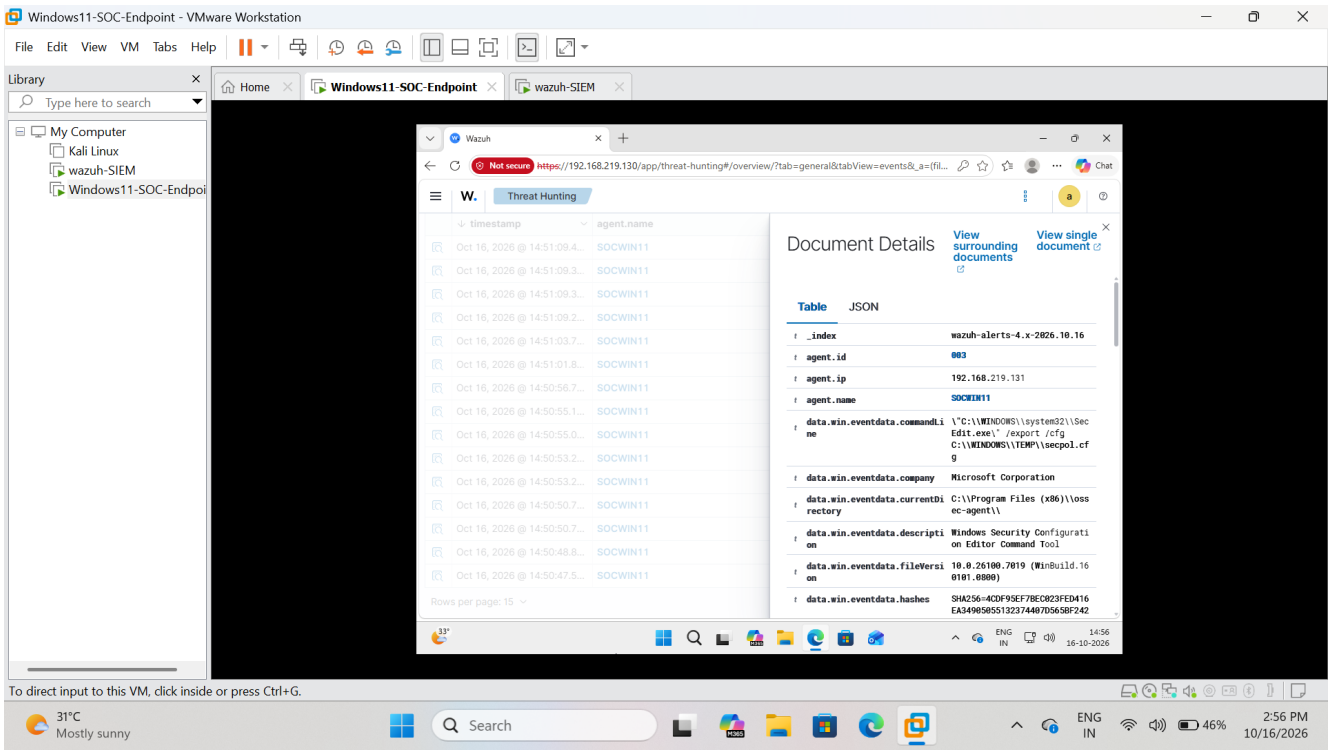


Figure 9. Wazuh detection details — Rule 92058 and MITRE T1546.011

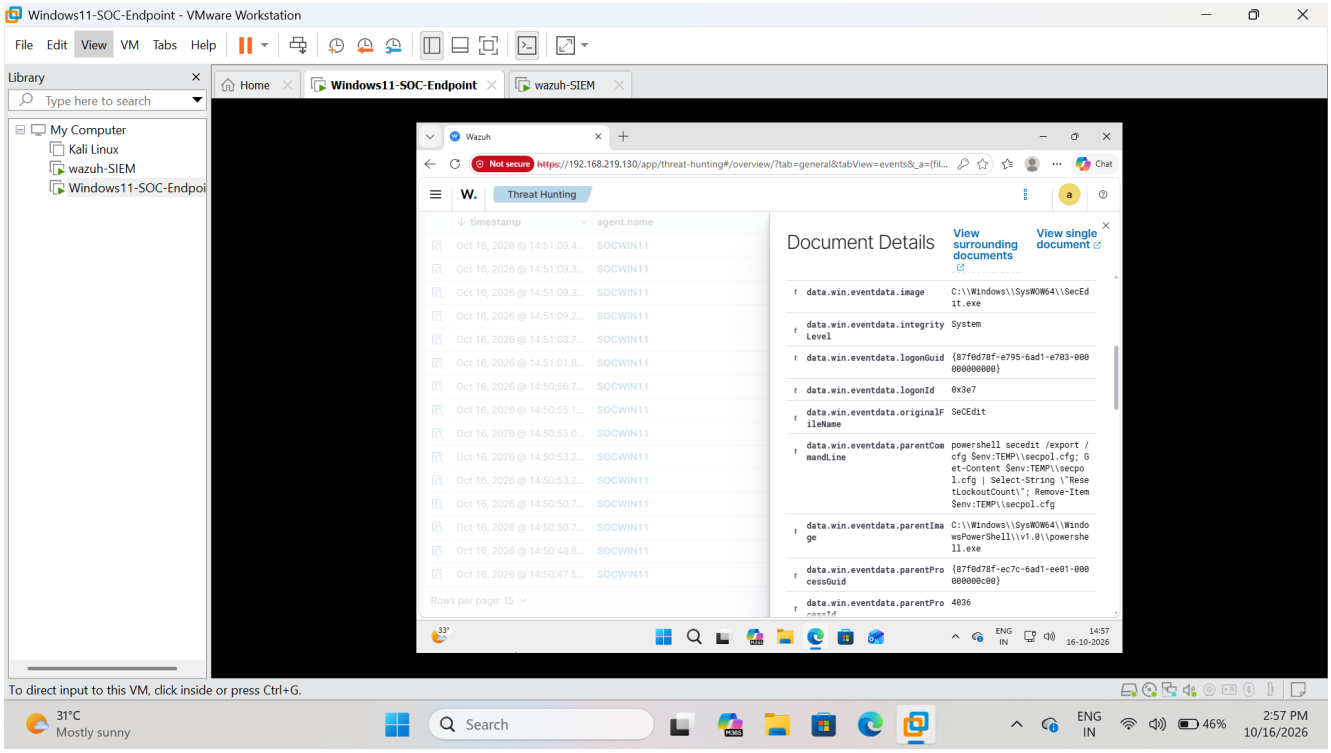


Figure 10. Wazuh event detail — event from the endpoint

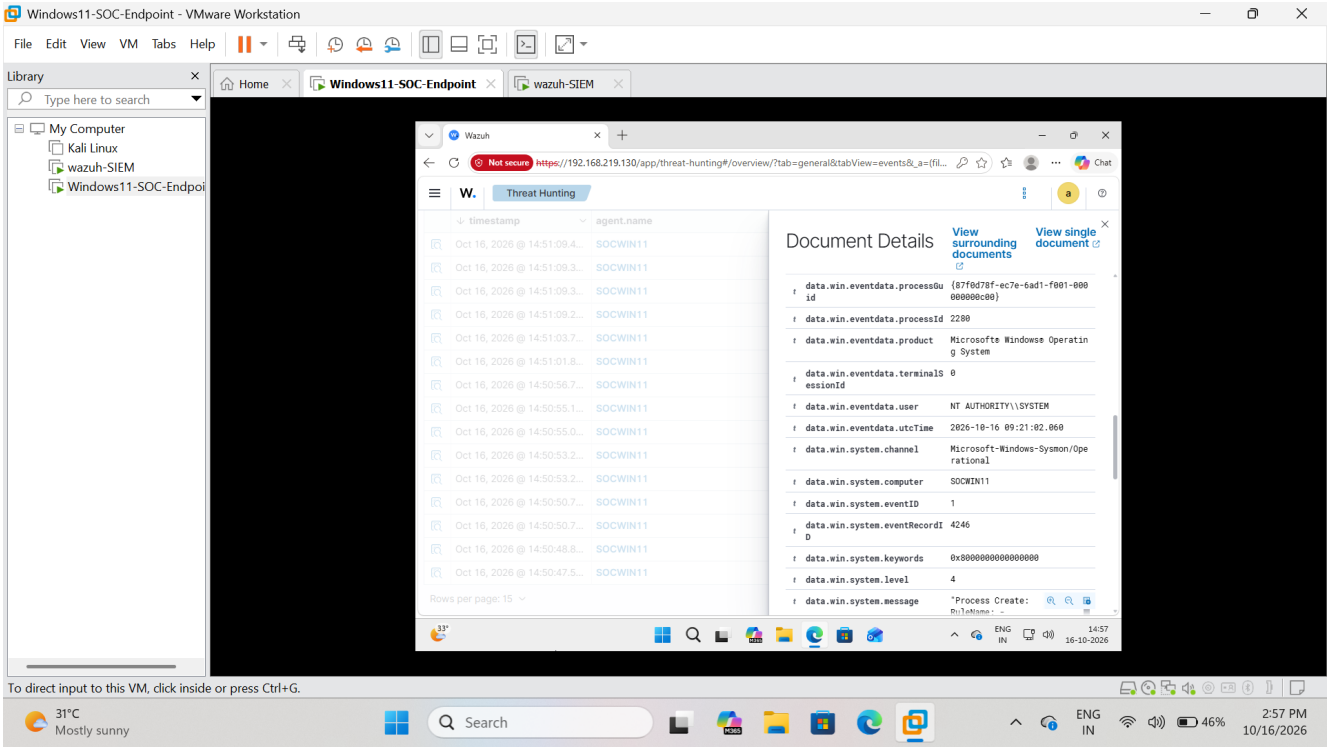


Figure 11. Wazuh event detail — event metadata

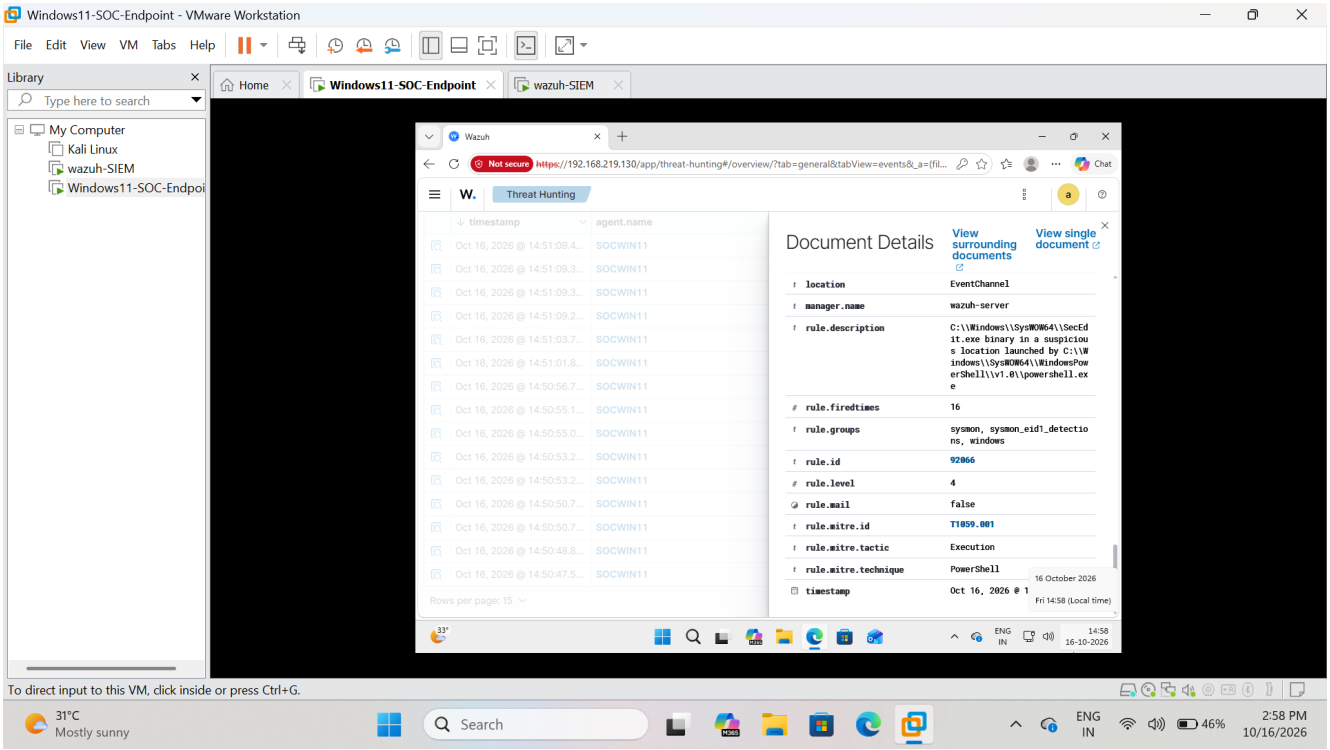


Figure 12. Wazuh event detail — additional rule context

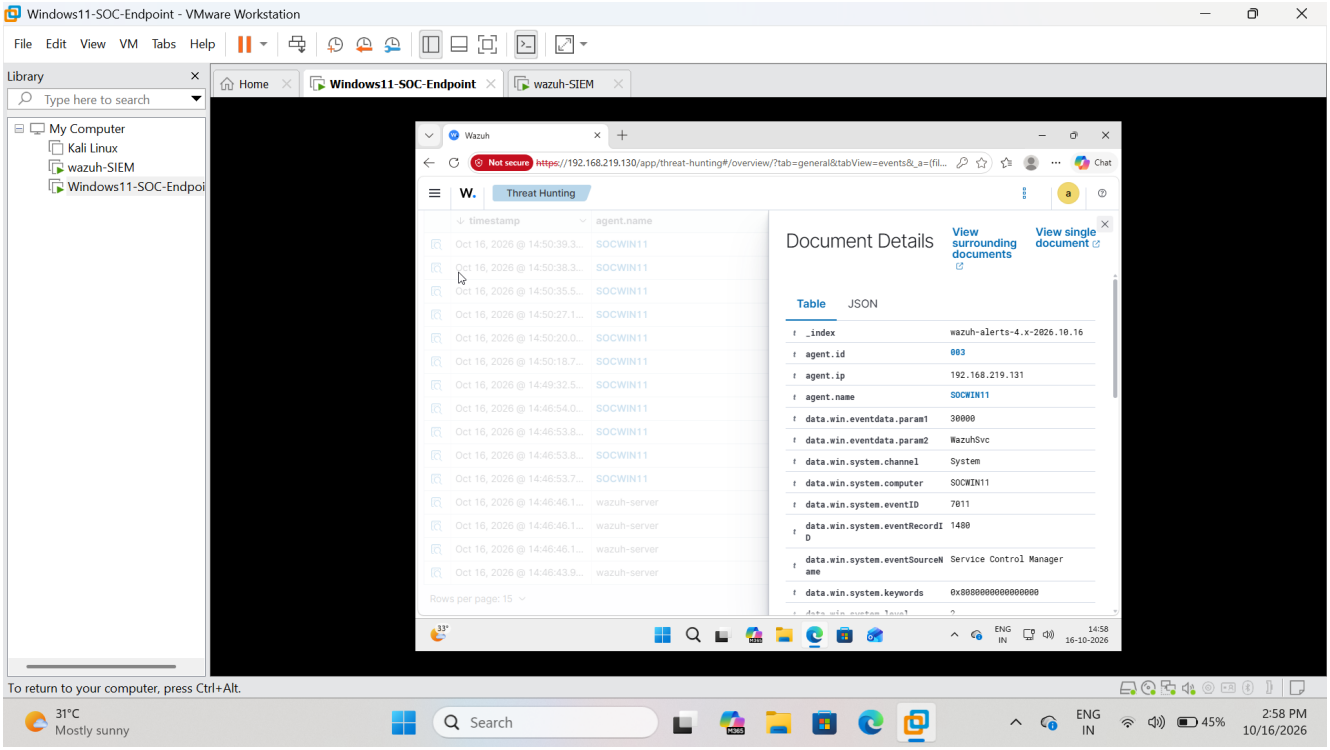


Figure 13. Wazuh event detail — rule and timestamp

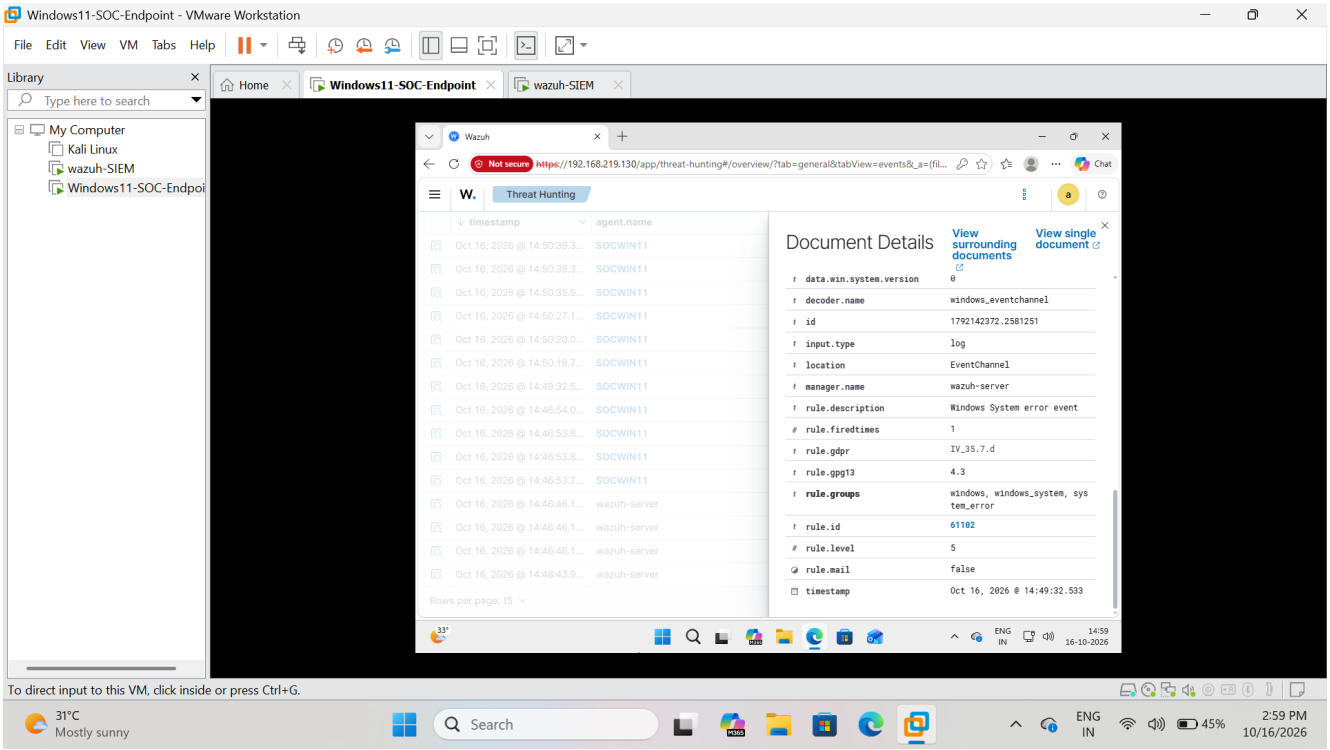


Figure 14. Wazuh event detail — SOCWIN11 agent information

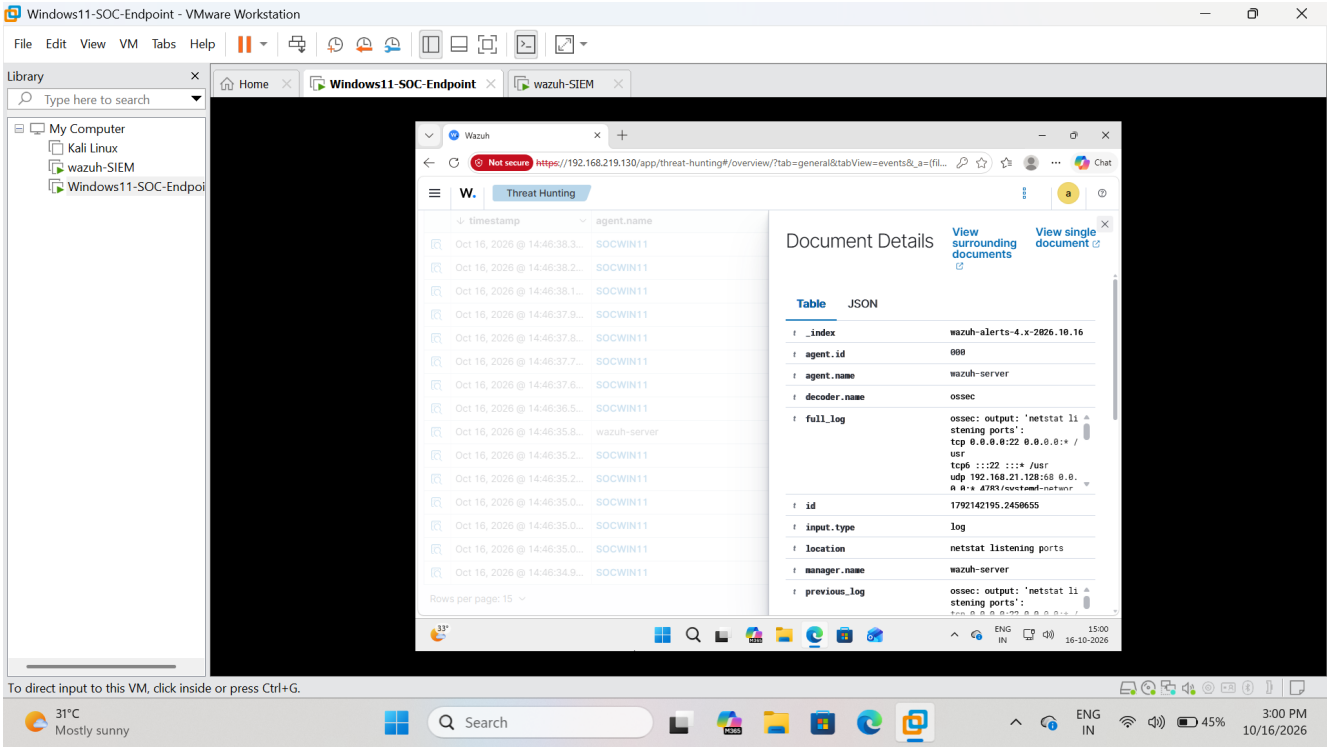


Figure 15. Wazuh event detail — Windows command line

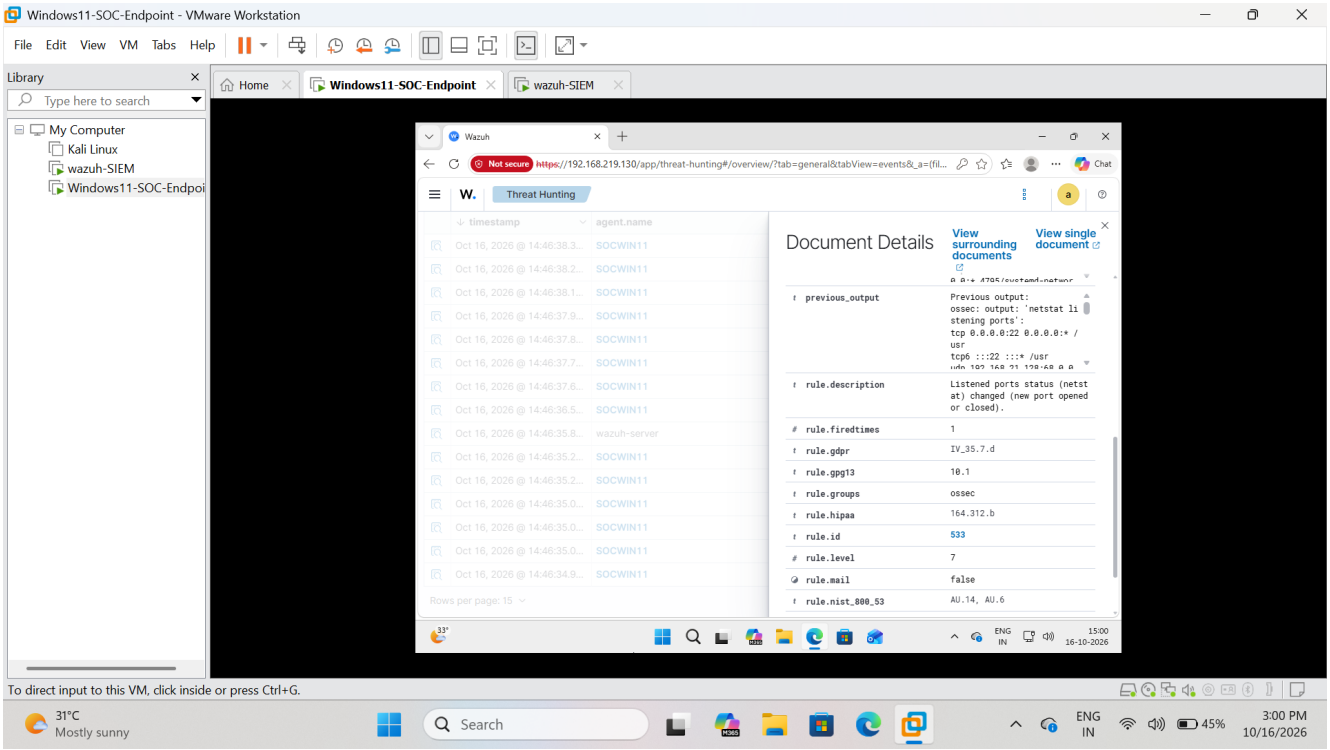


Figure 16. Wazuh event detail — Sysmon event fields

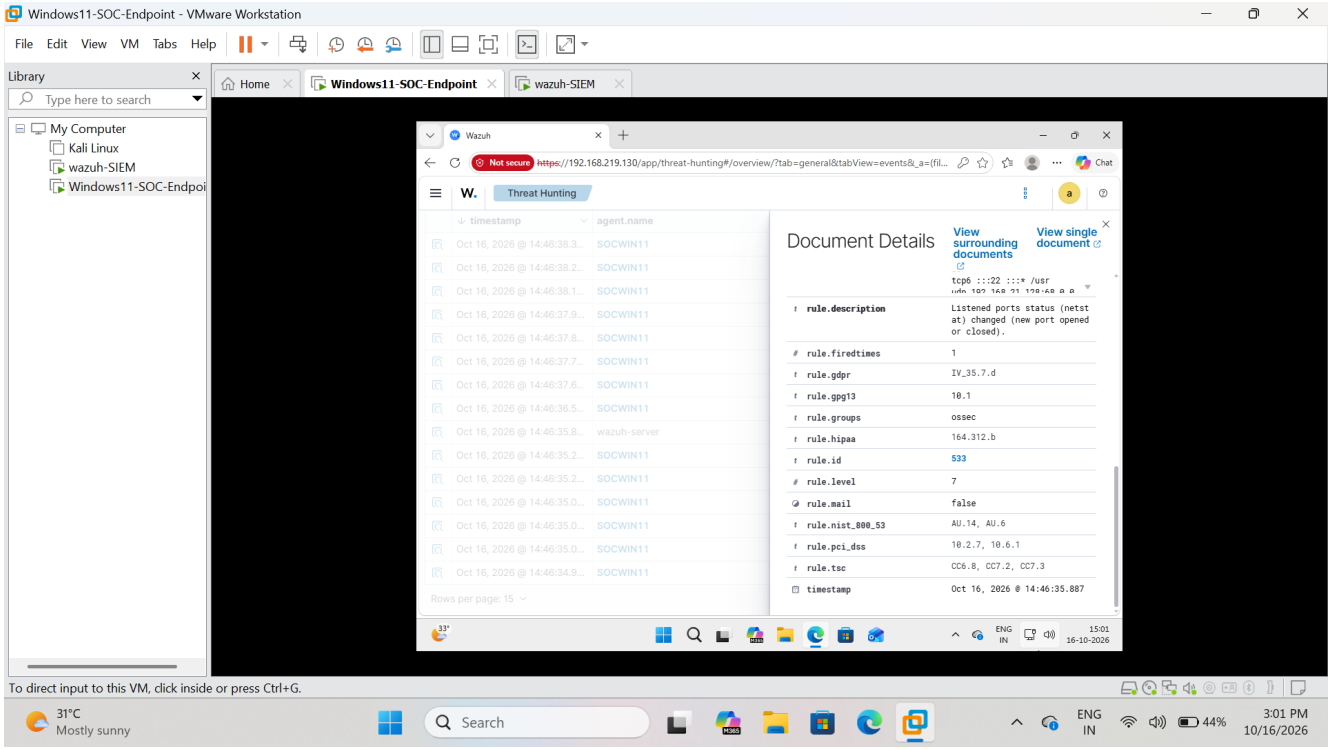


Figure 17. Wazuh event detail — additional event fields

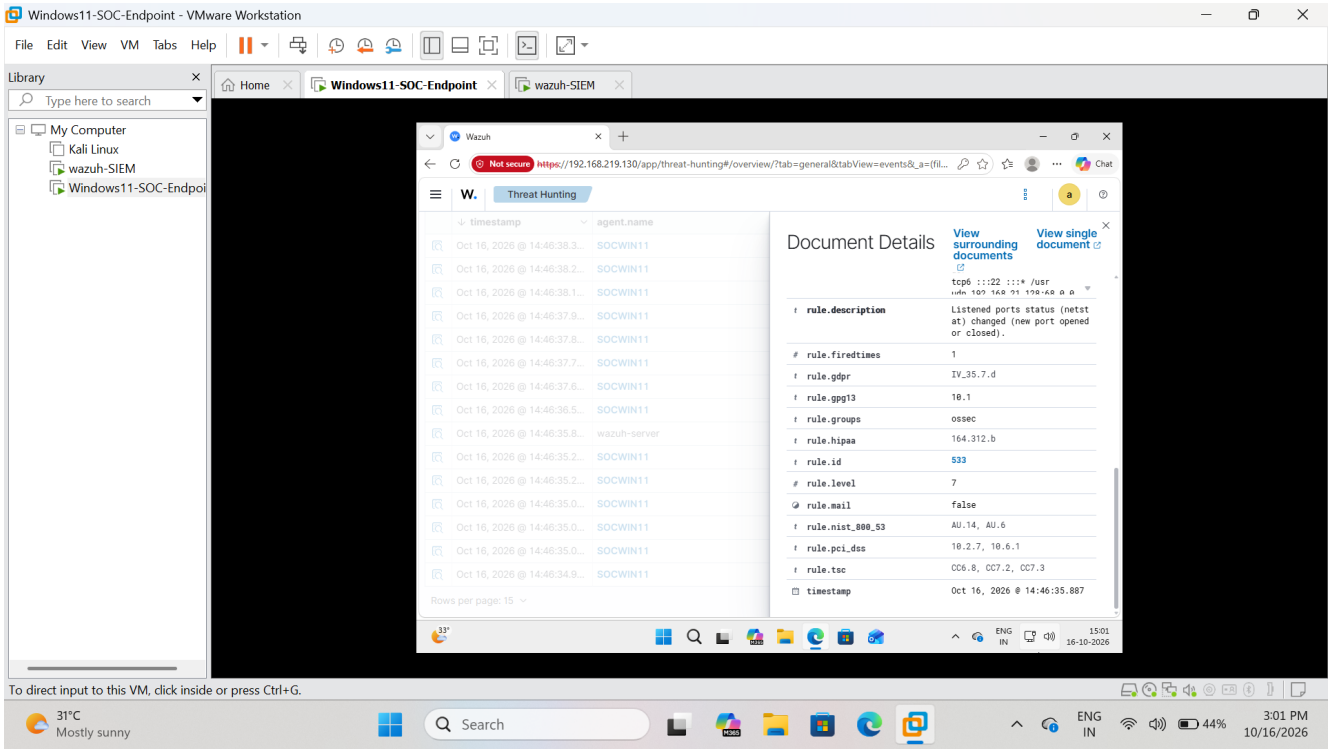


Figure 18. Wazuh event detail — MITRE ATT&CK; mapping

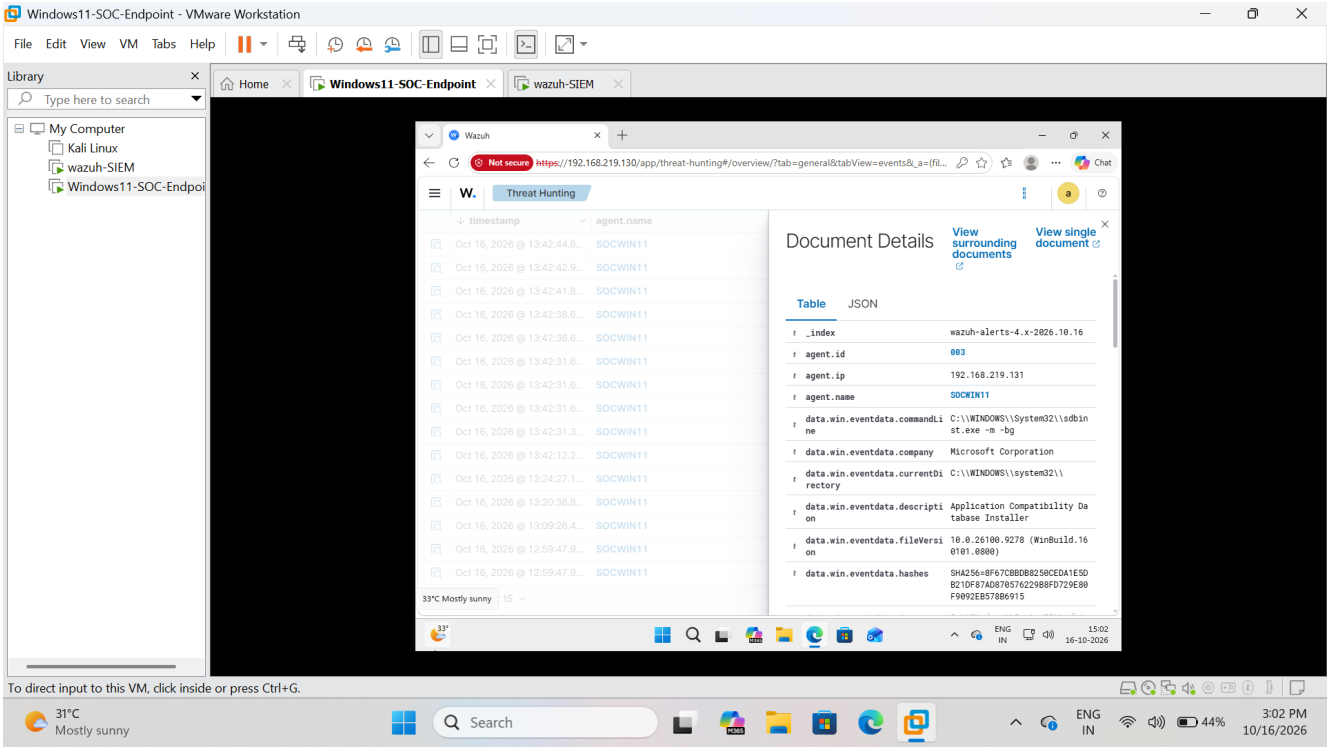


Figure 19. Wazuh event detail — application compatibility detection

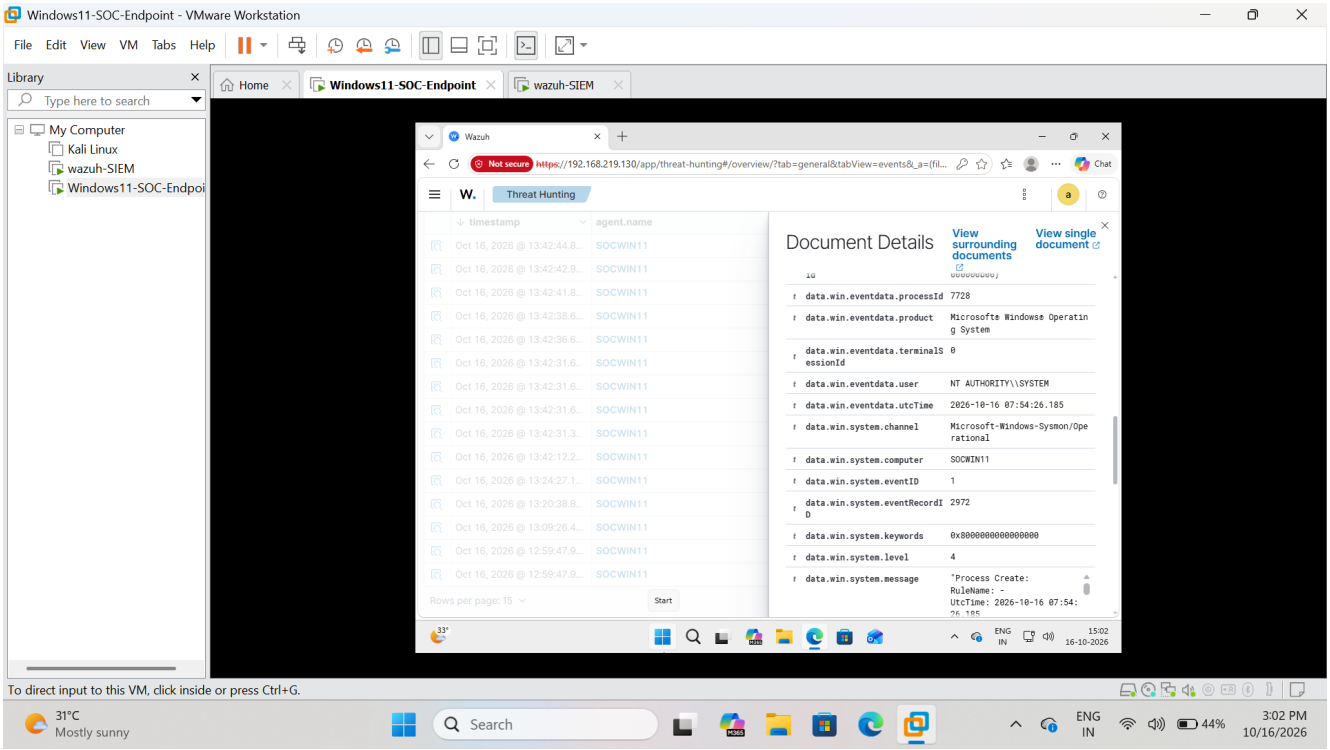


Figure 20. Wazuh event detail — Windows endpoint telemetry

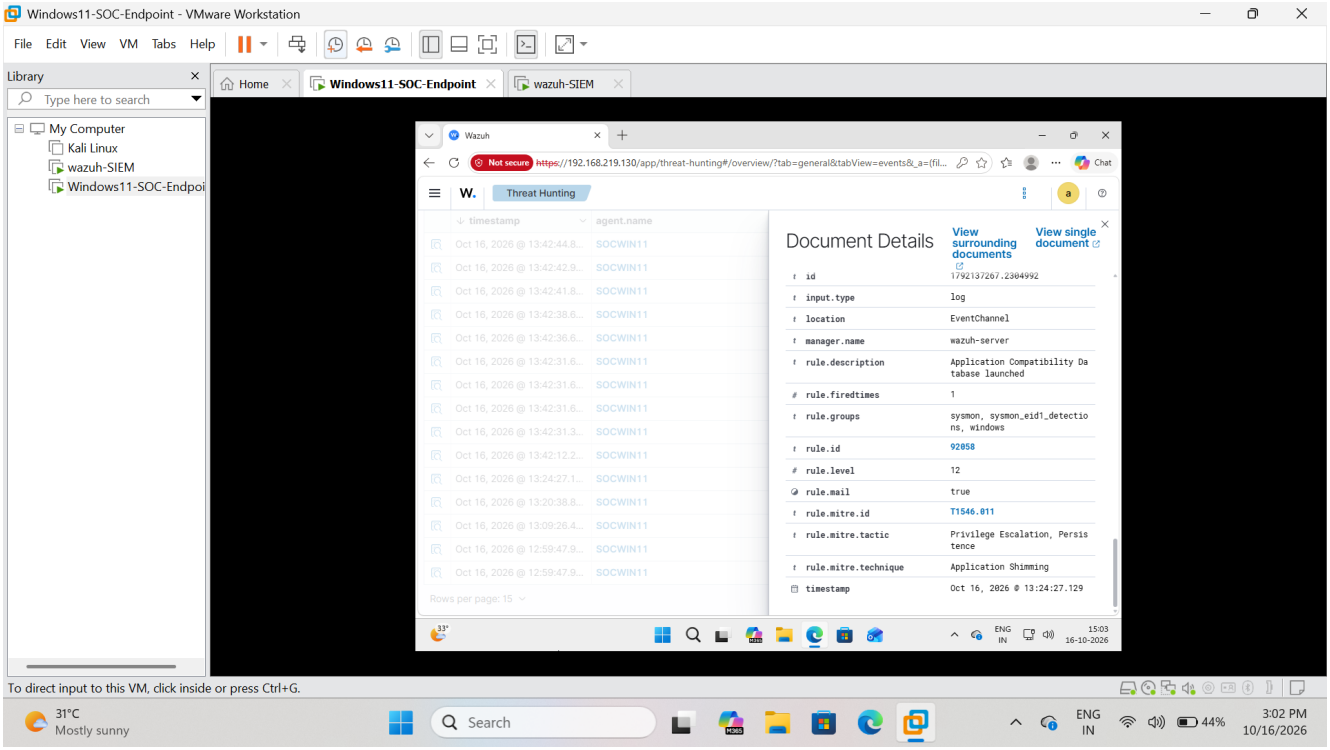


Figure 21. Wazuh event detail — event record and process data

6. Example alert investigation

Rule 92066 is visible in the supplied evidence. Its description flags SecEdit.exe running from a location described as suspicious and indicates PowerShell as the parent process. The event details show MITRE ATT&CK; T1059.001 (PowerShell), tactic Execution, and Wazuh rule level 4.

Initial interpretation

PowerShell and SecEdit.exe are legitimate Windows tools. Their presence is not automatically malicious. PowerShell launching a security configuration utility from an unusual path is worth reviewing because built-in tools can be misused. Treat the alert as a lead, not proof of compromise.

SOC L1 checks

• Confirm timestamp, endpoint, and agent ID.

• Review Image, CommandLine, ParentImage, ParentCommandLine, user, and process IDs.

• Check the executable path, signature, hash, and expected Windows location.

• Search nearby events for PowerShell activity, file creation, persistence, or security-setting changes.

• Confirm whether an administrator or lab action explains the behavior.

• Record evidence and escalate if unexplained.

Other visible detections

Rule 92031 — Level 3: “Discovery activity executed”; MITRE T1087 Account Discovery. Review process and command line to determine whether account enumeration was expected.

Rule 92066 — Level 4: SecEdit.exe in a suspicious location launched by PowerShell; MITRE T1059.001. Validate path, command line, process relationship, and context.

Rule 92058 — Level 12: “Application Compatibility Database Installer launched”; MITRE T1546.011 Application Shimming. Prioritize review, but validate context before concluding malicious activity.

7. SOC L1 workflow

• Triage the rule, severity, endpoint, and timestamp.

• Validate that the event belongs to SOCWIN11.

• Collect process, parent, command-line, user, hash, and surrounding-event context.

• Correlate related Sysmon and Windows events.

• Classify based on evidence and document uncertainty.

• Escalate high-severity or unexplained activity according to procedure.

8. Results, limitations, and next steps

Results supported by evidence

• Sysmon records Event IDs 1 and 5 locally.

• Wazuh Threat Hunting shows events associated with SOCWIN11.

• A Wazuh event document confirms Sysmon channel Event ID 1 forwarding.

• Wazuh event details expose telemetry, rule metadata, and MITRE mappings.

Limitations

• Screenshots show selected events from one lab session, not complete coverage of all Sysmon event types.

• An alert is not proof of compromise; command-line and process context must be checked.

• Dashboard hit totals can vary with time range and incoming events.

• The evidence does not establish that the captured detections were malicious.

Next steps

• Investigate the level 12 Rule 92058 first, then review Rules 92066 and 92031.

• Build a short timeline for each event and record the investigation conclusion.

• Keep the original screenshots and include a project README with setup, evidence, findings, and lessons learned.

9. Conclusion

The lab demonstrates the endpoint monitoring workflow: Sysmon records detailed process activity, the Wazuh agent forwards event-channel logs, and Wazuh Threat Hunting makes events searchable. The Sysmon Event ID 1 document in Wazuh verifies that Sysmon telemetry reached the SIEM. The main SOC lesson is to use process lineage, command lines, paths, hashes, user context, and related events before classifying an alert.

Appendix A — Key configuration

The Windows Wazuh agent configuration includes this event-channel block inside the ossec_config root element:

```
<localfile>
  <location>Microsoft-Windows-Sysmon/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
```

Configuration file: C:\Program Files (x86)\ossec-agent\ossec.conf. The block should appear before the closing </ossec_config> tag.

Glossary

SIEM: Security Information and Event Management platform for collecting and searching events.

Sysmon: Windows service that logs detailed activity such as process creation.

Wazuh agent: Endpoint component that forwards selected logs to Wazuh.

Event ID 1: Sysmon Process Create.

Event ID 5: Sysmon Process Terminated.

MITRE ATT&CK;: Knowledge base of adversary tactics and techniques.

Parent process: The process that started another process.

Additional Section — SOC Alert Case Studies and Timeline

This section summarizes three Wazuh detections visible in the supplied screenshots. These are triage exercises, not confirmed incidents. The available screenshots provide rule metadata and selected event fields, but do not establish that any activity was malicious.

Case Study 1: Rule 92066 — PowerShell associated with SecEdit.exe

Observed evidence: Wazuh rule 92066, reported rule level 4, associates PowerShell activity with SecEdit.exe. The visible command-line fragment begins with C:\Windows\SysWOW64\secE... and the event is mapped to MITRE ATT&CK; T1059.001 (PowerShell, Execution). The fragment is truncated, so the full command line and process lineage cannot be confirmed from the visible text.

Analyst interpretation: PowerShell and SecEdit.exe are legitimate Windows tools. An unusual process relationship or path can merit review, but the alert alone does not prove misuse.

Recommended checks: validate the complete image path, command line, parent image and parent command line, user, timestamp, signature/hash, and nearby process or security-configuration events.

Current disposition: Inconclusive; requires contextual validation.

Case Study 2: Rule 92058 — Application Compatibility Database Installer launched

Observed evidence: The screenshot shows rule 92058 at level 12 and a mapping to MITRE ATT&CK; T1546.011 (Application Shimming).

Analyst interpretation: Application compatibility mechanisms have legitimate uses and can also be abused. A high rule level warrants prompt review, but the rule match itself is not proof of persistence or compromise.

Recommended checks: validate the executable path and signature, parent process, command line, user, timestamp, relevant compatibility database or registry changes, and whether software installation or administration was expected.

Current disposition: Prioritize review; no confirmed compromise based on the screenshot alone.

Case Study 3: Rule 92031 — Discovery activity executed

Observed evidence: The screenshot shows rule 92031, level 3, described as “Discovery activity executed,” mapped to MITRE ATT&CK; T1087 (Account Discovery).

Analyst interpretation: Account discovery can be part of routine administration or attacker reconnaissance. The visible rule metadata does not identify enough context to classify the activity.

Recommended checks: identify the process and complete command line, user, timestamp, scope of account queries, and nearby events; compare with expected administrative activity.

Current disposition: Low-severity triage lead; requires context before classification.

Evidence-based timeline

Stage	Evidence observed	What it establishes
1. Endpoint telemetry	Sysmon Event ID 1 and Event ID 5 visible in Windows Event Viewer	Sysmon is recording process creation and termination locally.
2. Agent ingestion	Wazuh event details identify SOCWIN11, agent ID 003	Events are associated with the intended endpoint agent.
3. Sysmon forwarding	Wazuh document shows channel Microsoft-Windows-Sysmon/Operational and Event ID 1	At least one Sysmon process-creation event reached Wazuh.
4. Detection triage	Rules 92031, 92066, and 92058 visible in Wazuh screenshots	Wazuh produces detections and MITRE metadata for analyst review.
5. Disposition	Command-line and process context are incomplete in available evidence	No alert should be labeled malicious without further validation.

Final case-study conclusion

The lab demonstrates the end-to-end collection and initial triage of Windows endpoint telemetry. The strongest technical verification is the Wazuh event document that identifies the Sysmon Operational channel and Event ID 1 for SOCWIN11. The three alert examples demonstrate how rule metadata and MITRE ATT&CK; mappings guide investigation. The current evidence supports “detected and reviewed,” not “confirmed malicious.”